
(a) Symmetric page selections
selected in reference
reference page missed
selected outside reference
far (75% of mid tokens)
near
page mass
mass rank
avg pool
minmax
0
3000
6000
9000
12000
key position (tokens); query at right end
10^{-2}
10^{-3}
mass (log)
(b) Missed mass by region
missed near
missed far
missed reference mass
0.0
0.1
0.2
0.3
avg pool
minmax
(c) Which 32 of 128 dims
selected mass / reference mid-TopK mass
0.00
0.25
0.50
0.75
1.00
0.84
0.51
0.24
tail32 (low-freq)
random32 (unpaired)
high32 (high-freq)
(d) Historical far proxy (16 seqs)
gov_0
gov_1
hot_0
hot_1
mfld_0
mfld_1
mus_0
mus_1
nar_0
nar_1
pr_0
pr_1
qas_0
qas_1
natural
needle
0
8
16
24
32
layer
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
far proxy (log)